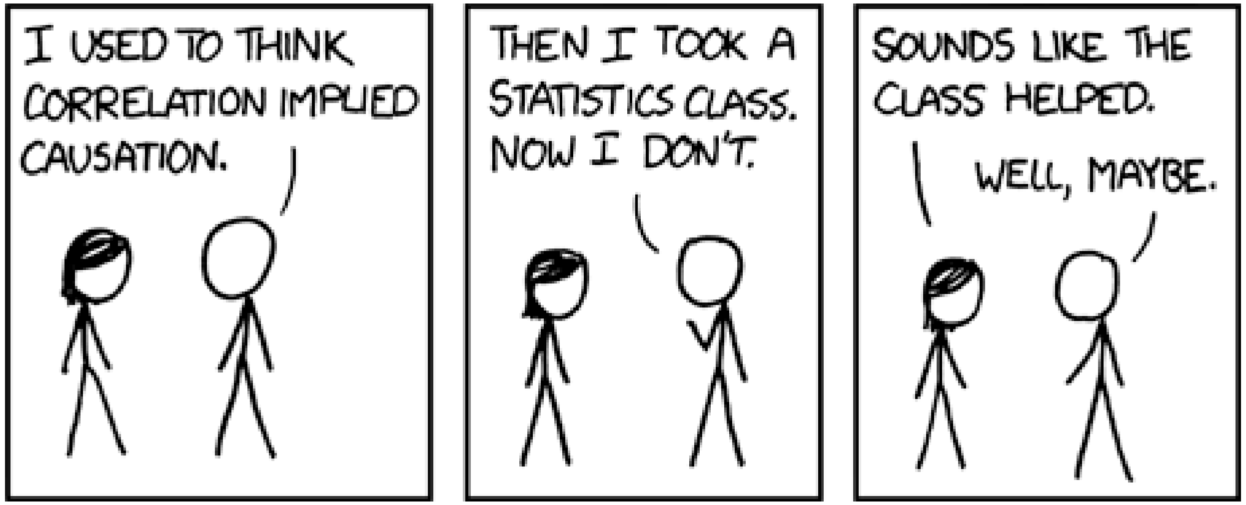
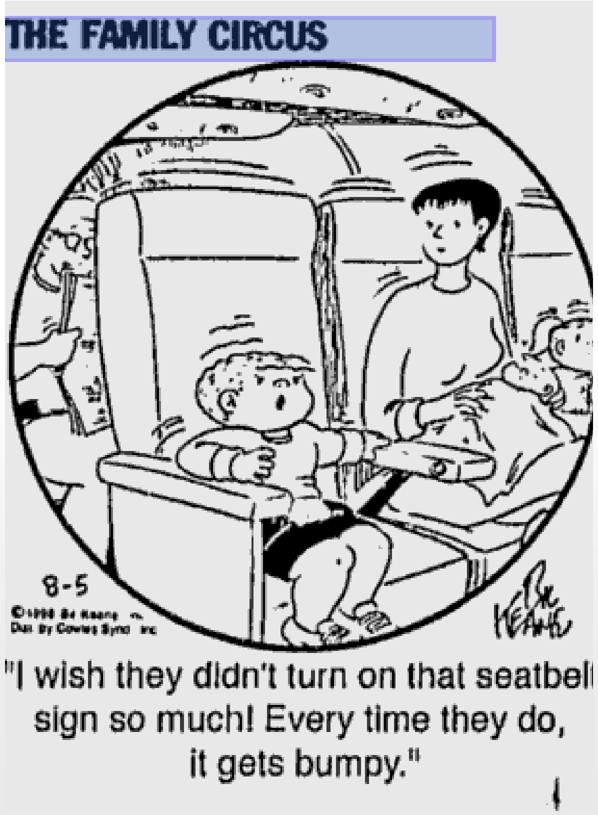


I USED TO THINK CORRELATION IMPLIED CAUSATION.
THEN I TOOK A STATISTICS CLASS. NOW I DON'T.
SOUNDS LIKE THE CLASS HELPED.
WELL, MAYBE.

CAUSATION

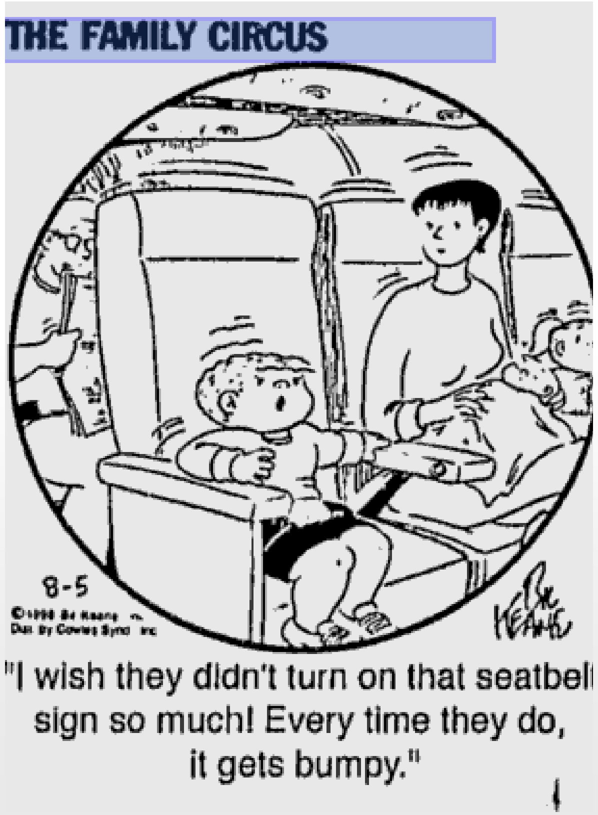
CORRELATION ⇝ CAUSATION

- Adage that "Correlation does not imply Causation" is a warning against
  - *Post hoc ergo propter hoc*
    - After this, therefore because of this
  - Specific case of Questionable Cause Fallacy
    - Logical Fallacy
    - Sometimes called
      - causal fallacy,
      - false cause, or
      - 'non causa pro causa'
        - Not the cause for the cause



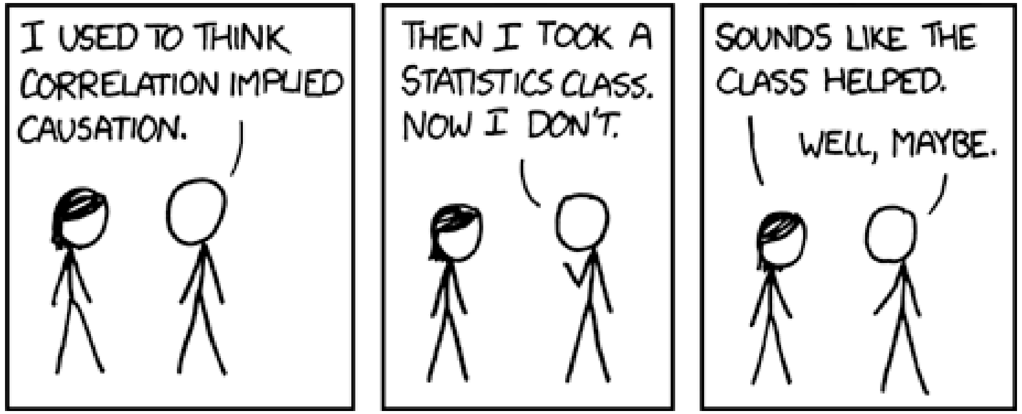
CORRELATION ⇝ CAUSATION

- We cannot assume that correlation means causation
  - A causes B; (direct causation)
  - B causes A; (reverse causation)
  - A and B are consequences of a common cause, but do not cause each other;
  - A causes B and B causes A (bidirectional or cyclic causation);
  - A causes C which causes B (indirect causation);
  - There is no connection between A and B; the correlation is a coincidence.



CORRELATION ⇝ CAUSATION

- “Correlation doesn't imply causation, but it does waggle its eyebrows suggestively and gesture furtively while mouthing *look over there.*”
  - https://xkcd.com/552/
- Correlations are necessary for causation, but not sufficient.



NECESSARY VERSUS SUFFICIENT (LOGIC)

Necessary (if *A*, then *B*; denoted $A \Rightarrow B$)

- The assertion that *B* is necessary for *A* is equivalent to
  - "*A* cannot be true unless *B* is true" or
    - "if B is false, then A is false".
  - (By contraposition, this is the same thing as "whenever *A* is true, so is *B*".)

Example:

- Being at least 30 years old is necessary for serving in the U.S. Senate.
- If you are under 30 years old, then it is impossible for you to be a senator.
- That is, if you are a senator, it follows that you are at least 30 years old.

NECESSARY VERSUS SUFFICIENT (LOGIC)

Sufficient

- If *X* is sufficient for *Y*, then knowing *X* to be true is means you can conclude that *Y* is true
  - however, knowing *X* to be false does not meet a minimal need to conclude that *Y* is false.

Example:

- If the U.S. Congress passes a bill, the president's signature of the bill is sufficient to make it law.
- But, if the president did not sign the bill, e.g. through exercising a presidential veto, that does not mean that the bill has not become law.
  - (it could still have become law through a congressional override).

CORRELATION ⇝ CAUSATION

Although correlation isn't sufficient, it is necessary component for causation.

Hill's Criteria of Causation

Austin Bradford Hill, "The Environment and Disease: Association or Causation?," Proceedings of the Royal Society of Medicine, 58 (1965): 295-300.

Cook and Campbell's Three Criteria (1979)

Cook, T. D. and Campbell, D. T.(1979). Quasi-experimentation: Design & analysis issues for field settings. Boston: Houghton Mifflin.

HILL'S CRITERIA

Strength:

A relationship is more likely to be causal if the correlation coefficient is large and statistically significant.

Consistency:

A relationship is more likely to be causal if it can be replicated.

HILL'S CRITERIA

Specificity:

A relationship is more likely to be causal if there is no other likely explanation.

Temporality:

A relationship is more likely to be causal if the effect always occurs after the cause.

HILL'S CRITERIA

Gradient:

A relationship is more likely to be causal if a greater exposure to the suspected cause leads to a greater effect.

Plausibility:

A relationship is more likely to be causal if there is a plausible mechanism between the cause and the effect.

HILL'S CRITERIA

Coherence: A relationship is more likely to be causal if it is compatible with related facts and theories.

Experiment: A relationship is more likely to be causal if it can be verified experimentally.

Analogy: A relationship is more likely to be causal if there are proven relationships between similar causes and effects.

HILL'S CRITERIA FULL LIST

- Strength
- Consistency
- Specificity
- Temporality
- Gradient
- Plausibility
- Coherence
- Experiment
- Analogy

HILLS CRITERIA IN PRACTICE: **SUBLUXATION**

Hill's Criteria of Causation Applied to Subluxation

	Criteria	Result
1	Strength	There were no studies that found a relative risk or odds ratio linking subluxation
2	Consistency	Subluxation has not been noted to be consistently found across any studies in different people, places, circumstances or time.
3	Specificity	There were no studies that linked disease with subluxation of any specificity. Other exposures (variables) or explanations can be given to the disease complex.
4	Temporal sequence	There were no studies suggestive of a temporal sequence linking subluxation with disease
5	Dose response	There were no studies found linking incidence of disease with magnitude of the subluxation
6	Experimental evidence	There were no consistent studies demonstrating subluxation in the animal model
7	Biological plausibility	No studies were found that offered reproducible evidence to suggest a biological plausibility of the subluxation construct.
8	Coherence	There were no studies that indicated a credible level of coherence
9	Analogy	There were no studies suggestive of a casual association via a similar agent.

- **Subluxation**
  - An orthopedic **subluxation**, a true vertebral misalignment, or a mechanical joint dysfunction that affects mobility in the spine is
  - not the same as a "**chiropractic subluxation**" that is alleged to cause disease by interfering with nerve supply to organs.

- Mirtz, T. A., Morgan, L., Wyatt, L. H., & Greene, L. (2009). An epidemiological examination of the subluxation construct using Hill's criteria of causation. *Chiropractic & osteopathy*, *17*(1), 13.
  - https://chiromt.biomedcentral.com/articles/10.1186/1746-1340-17-13

COOK AND CAMPBELL (1979)

Covariation:

- Changes in the assumed cause (X) are related to changes in the assumed effect (Y). Changing X results in a predictable change in Y.

Temporal Precedence:

- The assumed cause must occur before the assumed effect.

No Plausible Alternative Explanations:

- The assumed cause must be the only reasonable explanation for changes in the measured assumed effect.

NO PLAUSIBLE ALTERNATIVE EXPLANATIONS

The third criterion is the hardest to satisfy

Plausible alternative explanations are "threats to validity"

Goal is to Minimizing Threats to Validity

- Multiple approaches (can be combined)
  - Argument
  - Measurement or Observation
  - Analysis
  - Preventive Action
  - Design

GOAL IS TO MINIMIZING THREATS TO VALIDITY

- The five categories listed should not be considered mutually exclusive.
- The inclusion of measurements designed to minimize threats to validity will obviously be related to the design structure and is likely to be a factor in the analysis.
- A good research plan should, where possible,
  - make use of multiple methods for reducing threats.
- In general, reducing a particular threat by design or preventive action will probably be stronger than by using one of the other three approaches.
- The choice of which strategy to use for any particular threat is complex and
  - depends at least on the cost of the strategy and on the potential seriousness of the threat.

BY ARGUMENT

- The most straightforward way to rule out a potential threat to validity is to
  - argue that the threat in question is not a reasonable one.
- Such an argument may be made either *a priori* or *a posteriori*,
  - A priori is more convincing because you make the argument before (prior) to testing your results
  - *a posteriori* is less convincing because you make the argument after (post) testing your results

BY ARGUMENT

- For example,
  - an instrumentation threat is not likely because the same test is used for pre and post test measurements
  - and did not involve observers who might improve, or other such factors.
- Ruling out a potential threat to validity by argument alone will be weaker than the other approaches
- As a result, the most plausible threats in a study should not, except in unusual cases, be ruled out by argument only.

BY MEASUREMENT OR OBSERVATION

- In some cases, it will be possible to rule out a threat by measuring it
  - and demonstrating that either
    - it does not occur at all or
    - occurs so minimally
  - as to not be a strong alternative explanation for the cause-effect relationship.

BY MEASUREMENT OR OBSERVATION

- Consider, for example, a study of the effects of an advertising campaign on subsequent sales of a particular product.
  - In such a study, history (i.e., the occurrence of other events which might lead to an increased desire to purchase the product) would be a plausible alternative explanation.
  - For example, a change in the local economy, the removal of a competing product from the market, or similar events could cause an increase in product sales.
- One might attempt to minimize such threats by measuring local economic indicators and the availability and sales of competing products.
  - If there is no change in these measures coincident with the onset of the advertising campaign,
  - these threats would be considerably minimized.

BY ANALYSIS

- There are a number of ways to rule out alternative explanations using statistical analysis.
- One interesting example is provided by Jurs and Glass (1971).
  - They suggest that one could study the plausibility of an attrition or mortality threat by conducting an analysis of variance with two factors.
  - One factor in this study would be the original treatment group designations (i.e., program vs. comparison group), while the other factor would be attrition (i.e., dropout vs. non-dropout group).
  - The dependent measure could be the pretest or other available pre-program measures. A main effect on the attrition factor would be indicative of a threat to external validity or generalizability, while an interaction between group and attrition factors would point to a possible threat to internal validity. Where both effects occur, it is reasonable to infer that there is a threat to both internal and external validity.

BY ANALYSIS

- The plausibility of alternative explanations might also be minimized using analysis of covariance (ANCOVA).
- For example, in a study of the effects of "workfare" programs on social welfare case loads,
  - one plausible alternative explanation might be the status of local economic conditions.
  - Here, it might be possible to construct a measure of economic conditions and include that measure as a covariate in the statistical analysis.
  - One must be careful when using covariance adjustments of this type -- "perfect" covariates do not exist in most social research and the use of imperfect covariates will not completely adjust for potential alternative explanations.
  - Nevertheless causal assertions are likely to be strengthened by demonstrating that treatment effects occur even after adjusting on a number of good covariates.

BY PREVENTIVE ACTION

When potential threats are anticipated they can often be ruled out by some type of preventive action.

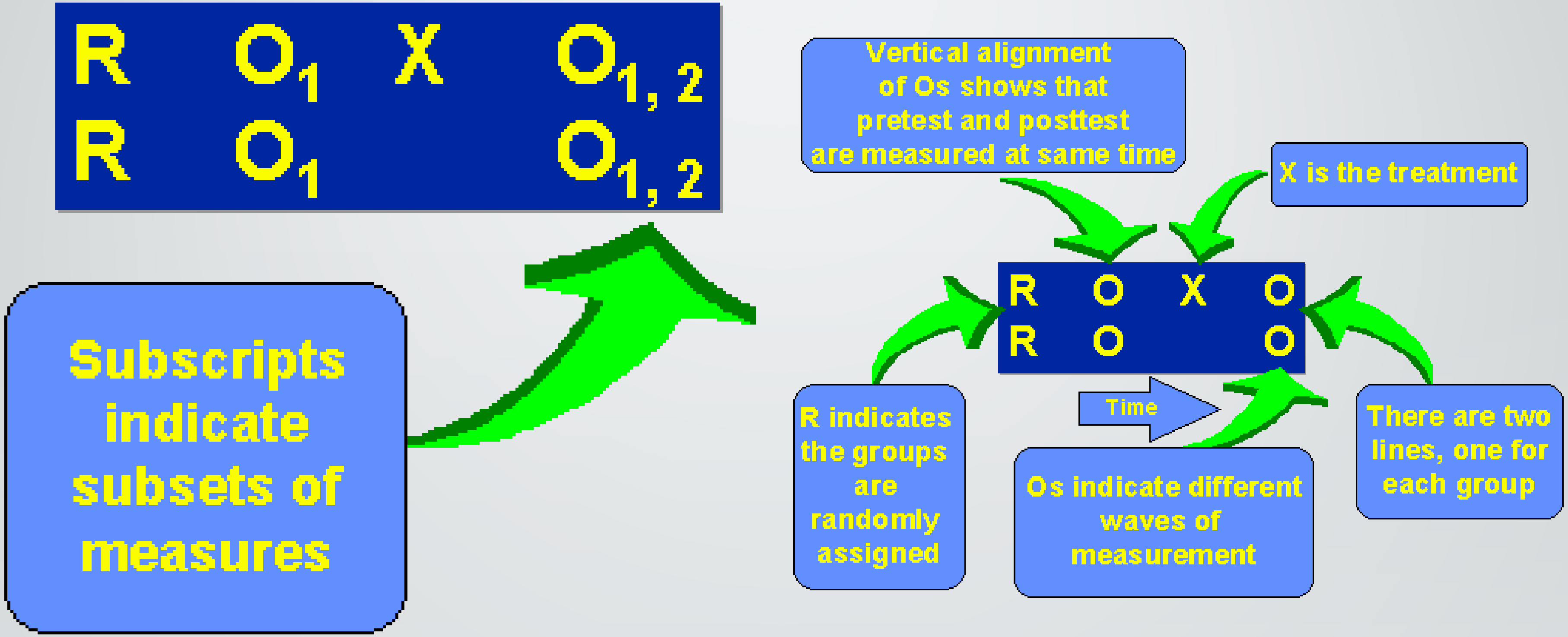
In addition, auditing methods and quality control can be used to track potential experimental dropouts or to insure the standardization of measurement.

BY PREVENTIVE ACTION

- For example, if the program is a desirable one, it is likely that the comparison group would feel jealous or demoralized.
- Several actions can be taken to minimize the effects of these attitudes including offering the program to the comparison group upon completion of the study or using program and comparison groups which have little opportunity for contact and communication.

BY DESIGN

- Here, the major emphasis is on ruling out alternative explanations by adding treatment or control groups, waves of measurement, and the like.
- Is the largest area of emphasis, so we can spend a lot of time on it
- Basic Design Elements. Most research designs can be constructed from four basic elements:
  - Time.
  - Program(s) or Treatment(s).
  - Observation(s) or Measure(s).
  - Groups or Individuals.



DESIGN CONSTRUCTION

TIME (PT 1)

- A causal relationship, by its very nature, implies that some time has elapsed between the occurrence of the cause and the consequent effect.
- While for some phenomena the elapsed time might be measured in microseconds and therefore might be unnoticeable to a casual observer,
  - we normally assume that the cause and effect in social science arenas do not occur simultaneously,

TIME (PT 2)

- In design notation we indicate this temporal element horizontally - whatever symbol is used to indicate the presumed cause would be placed to the left of the symbol indicating measurement of the effect.
- Thus, as we read from left to right in design notation we are reading across time.
- Complex designs might involve a lengthy sequence of observations and programs or treatments across time.

PROGRAM(S) OR TREATMENT(S).

- The presumed cause may be a
  - program or treatment under the explicit control of the researcher
  - or the occurrence of some natural event or program not explicitly controlled.
- In design notation we usually depict a presumed cause with the symbol "X".
- When multiple programs or treatments are being studied using the same design,
  - we can keep the programs distinct by using subscripts such as "X_1" or "X_2".
- For a comparison group (i.e., one which does not receive the program under study) no "X" is used.

OBSERVATION(S) OR MEASURE(S)

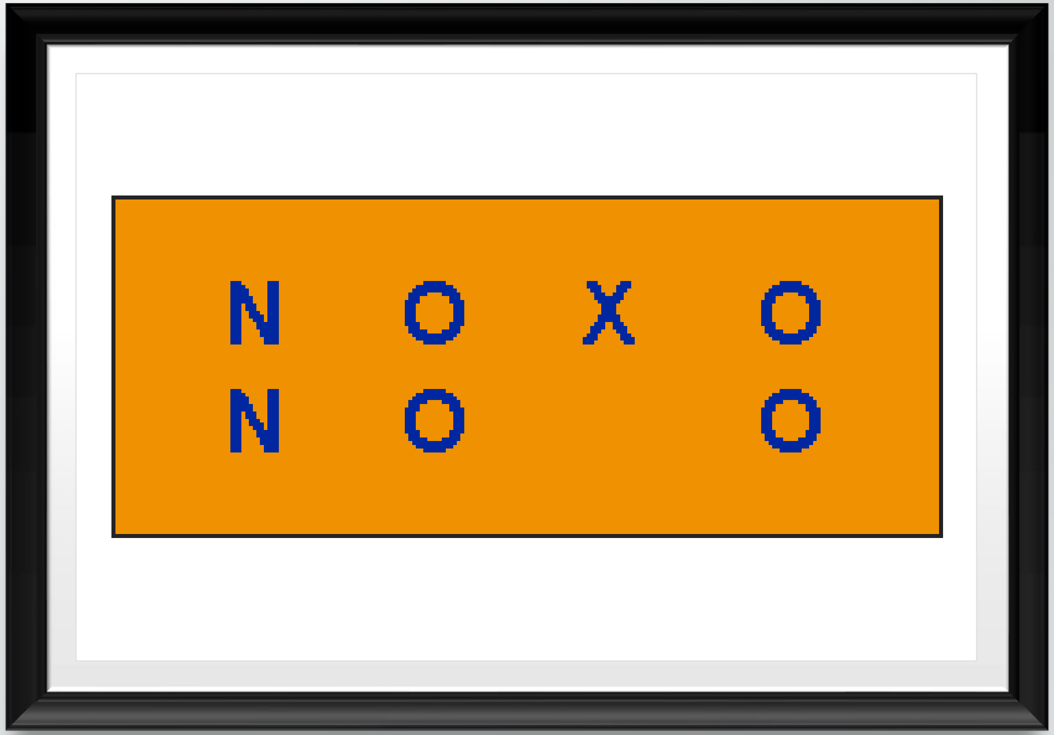
- Measurements are typically depicted in design notation with the symbol "O".
- If the same measurement or observation is taken at every point in time in a design,
    - then this "O" will be sufficient.
- Similarly, if the same set of measures is given at every point in time in this study,
    - the "O" can be used to depict the entire set of measures.
- However, if different measures are given at different times,
    - it is useful to subscript the "O" to indicate which measurement is being given at which point in time.

GROUPS OR INDIVIDUALS

- The final design element consists of the:
  - intact groups or the
  - individuals who participate in various conditions.
- Typically, there will be one or more program and comparison groups.
- In design notation, each group is indicated on a separate line.
- Furthermore, the way groups are assigned to the conditions can be indicated by an appropriate symbol at the beginning of each line.
- Here, "R" will represent a group which was randomly assigned,
  - "N" will depict a group which was nonrandomly assigned (i.e., a nonequivalent group or cohort) and
  - a "C" will indicate that the group was assigned using a cutoff score on a measurement.

ILLUSTRATIVE EXAMPLE
(TWO-GROUP PRE-POST DESIGN)

- Perhaps the easiest way to understand how
    - these four basic elements become integrated into a design structure is by example
- One of the most used designs in social research is the
    - two-group pre-post design which can be depicted as:



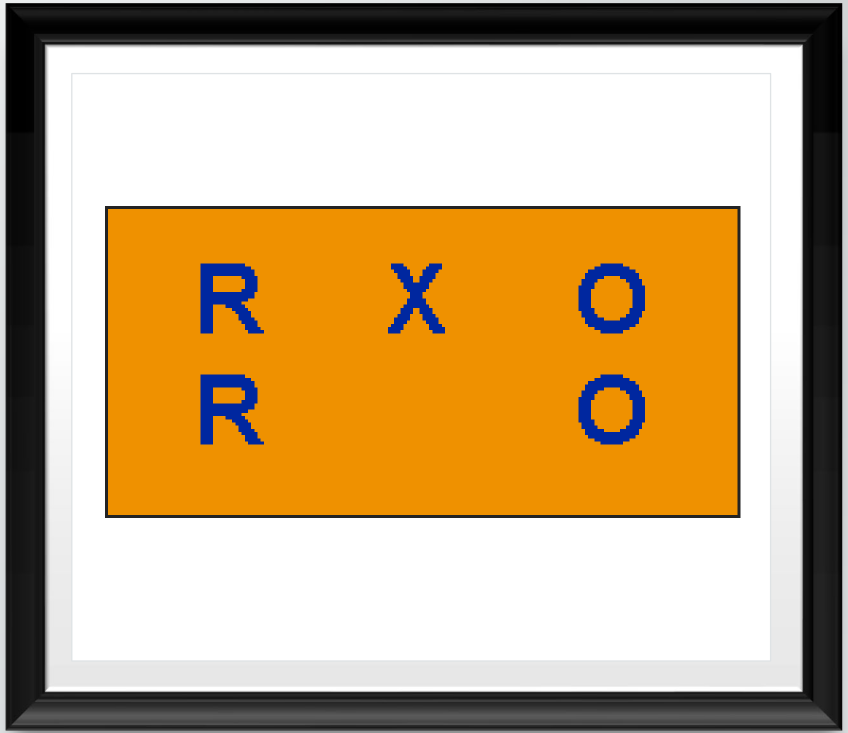
TWO-GROUP PRE-POST DESIGN

- There are two lines in the design indicating that the study was comprised of two groups.
- The two groups were nonrandomly assigned as indicated by the "N".
- Both groups were measured before the program or treatment occurred as indicated by the first "O" in each line.
- Following this preobservation,
  - the group in the first line received a program or treatment while the group in the second line did not.
- Finally, both groups were measured subsequent to the program.

N	O	X	O
N	O		O

POSTTEST-ONLY RANDOMIZED EXPERIMENT

- Another common design is the posttest-only randomized experiment.
- The design can be depicted as ->
- Here, two groups are randomly selected with
  - one group receiving the program and
  - one acting as a comparison.
- Both groups are measured after the program is administered.



EXPANDING A DESIGN

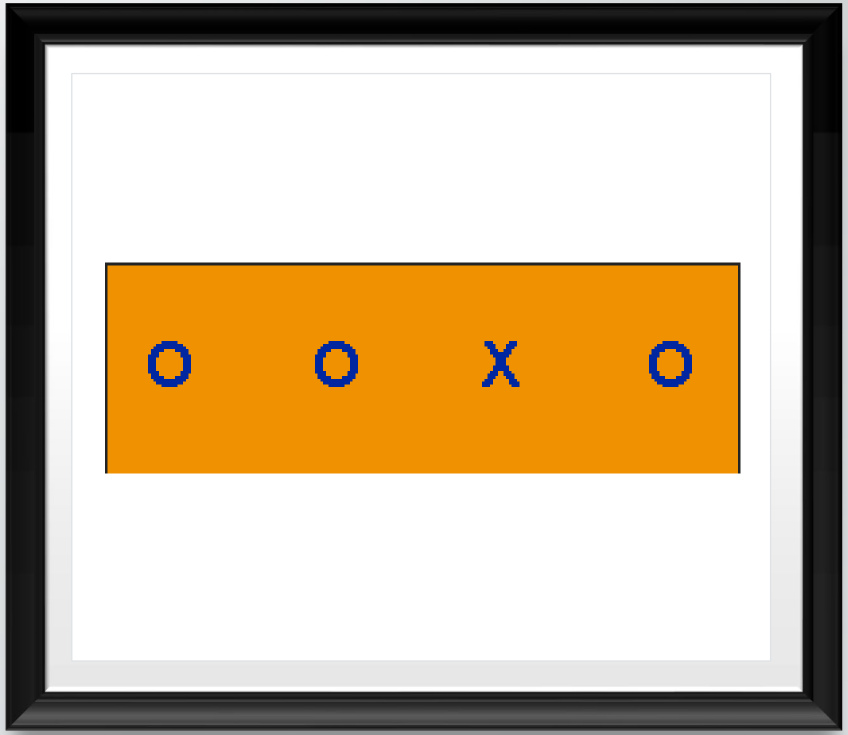
- We can combine the four basic design elements in several ways to arrive at a specific design,
  - which is appropriate for the setting at hand.
- One strategy for doing so begins with the basic causal relationship

EXPANDING A DESIGN

- This is the simplest design in causal research and serves as a
  - starting point for the development of better strategies.
- When we add to this basic design, we are essentially expanding one of the four basic elements described earlier.
- Each possible expansion has implications,
  - both for the cost of the study and
  - for the threats which might be ruled out.

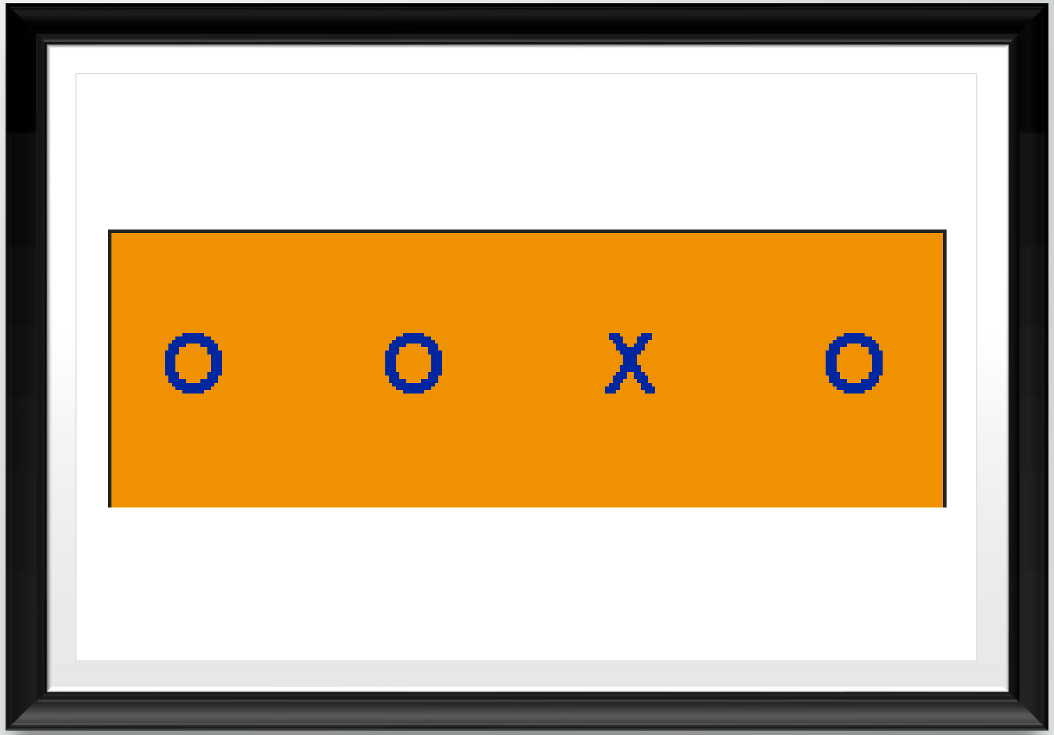
EXPANDING ACROSS TIME

- We can add to the basic design by including additional observations either
  - before or after the program or,
  - by adding or removing the program or different programs.
- For example, we might add one or more pre-program measurements and achieve the following design:



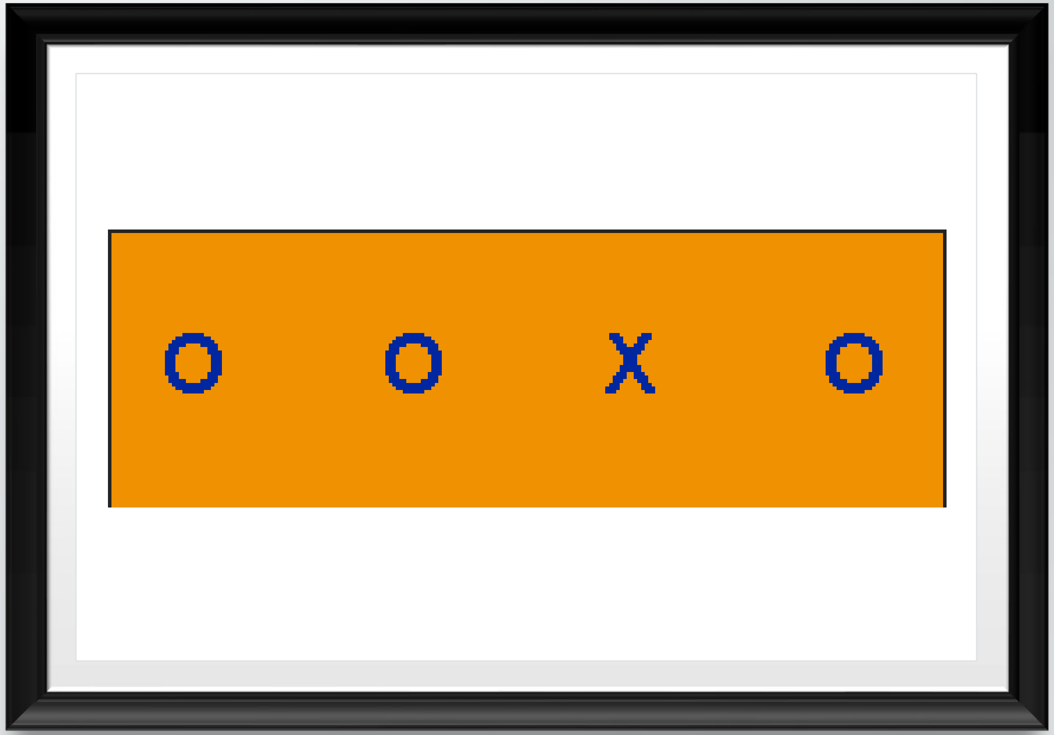
EXPANDING ACROSS TIME

- The addition of such pretests provides a "baseline" which, for instance,
  - helps to assess the potential of a maturation or testing threat.
- If a change occurs between the first and second pre-program measures,
  - it is reasonable to expect that similar change might be seen between the second pretest and the posttest,
  - even in the absence of the program.



EXPANDING ACROSS TIME

- However, if no change occurs between the two pretests,
  - one might be more confident in assuming that maturation or testing
  - is not a likely alternative explanation for the cause-effect relationship which was hypothesized.



EXPANDING ACROSS TIME

- Similarly, additional post-program measures could be added.
- This would be useful for determining whether an immediate program effect decays over time, or
  - whether there is a lag in time between the initiation of the program and the occurrence of an effect.
- We might also add and remove the program over time:

O X O O X O

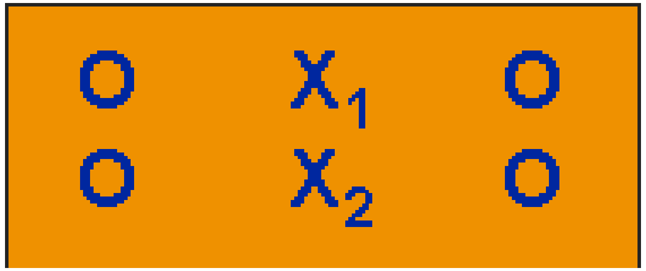
EXPANDING ACROSS TIME

- This is one form of the ABAB design which is frequently used in clinical psychology and psychiatry.
- The design is particularly strong against a history threat.
- When the program is repeated it is less likely that unique historical events can be responsible for replicated outcome patterns.

O X O O X O

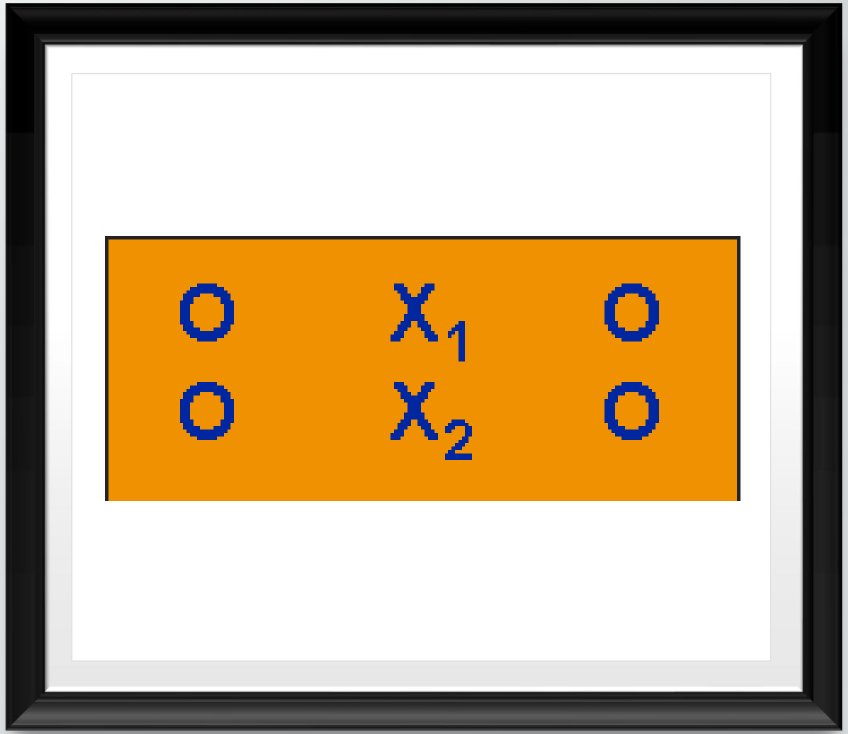
EXPANDING ACROSS PROGRAMS

- Another way to expand the program would be to partition it into different levels of treatment.
- For example, in a study of the effect of a novel drug on subsequent behavior,
  - we might use more than one dosage of the drug:



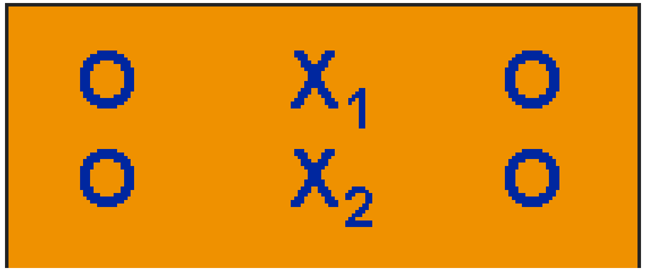
EXPANDING ACROSS PROGRAMS

- This design is an example of a simple factorial design with one factor having two levels.
- Notice that group assignment is not specified indicating that any type of assignment might have been used.
- This is a common strategy in a "sensitivity" or "parametric" study where the primary focus is one the effects obtained at various program levels.



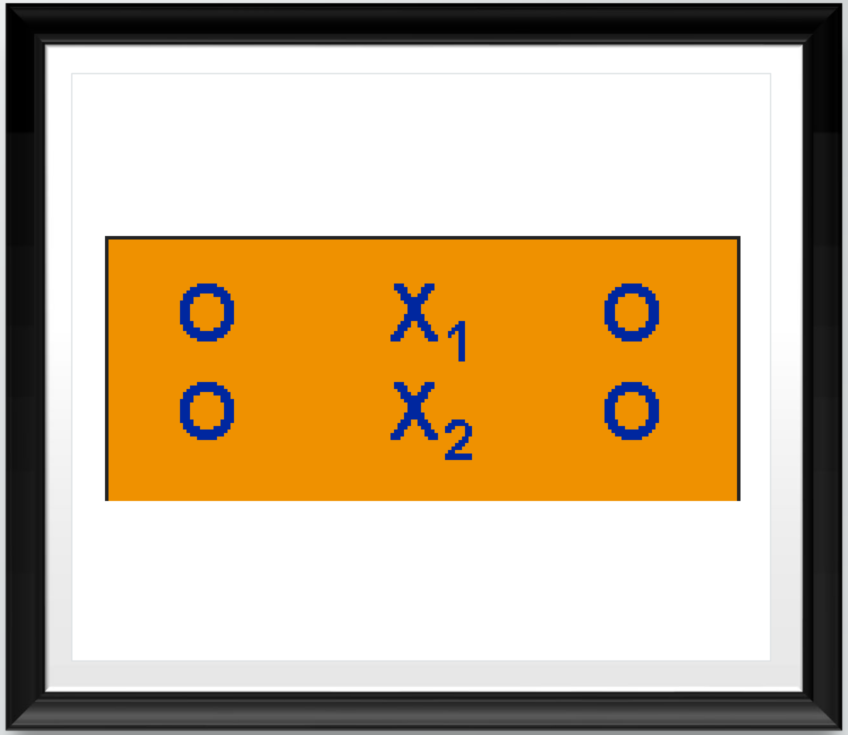
EXPANDING ACROSS PROGRAMS

- In a similar manner, one might expand the program by varying specific components of it across groups.
- This might be useful if one wishes to study different modes of the delivery of the program, different sets of program materials and the like.



EXPANDING ACROSS PROGRAMS

- Finally, we can expand the program by using theoretically polarized or "opposite" treatments.
- A comparison group can be considered one example of such a polarization.
- Another might involve use of a second program which is expected to have an opposite effect on the outcome measures.
- A strategy of this sort provides evidence that the outcome measure is sensitive enough to differentiate between different programs.



EXPANDING ACROSS OBSERVATIONS

- At any point in time in a research design it is usually desirable to collect multiple measurements.
- For example, we might add a number of similar measures in order to determine whether the results of these converge.
- Or, we might wish to add measurements which theoretically should not be affected by the program in question in order to demonstrate that the program discriminates between effects.
- Strategies of this type are useful for achieving convergent and discriminant validity.

EXPANDING ACROSS OBSERVATIONS

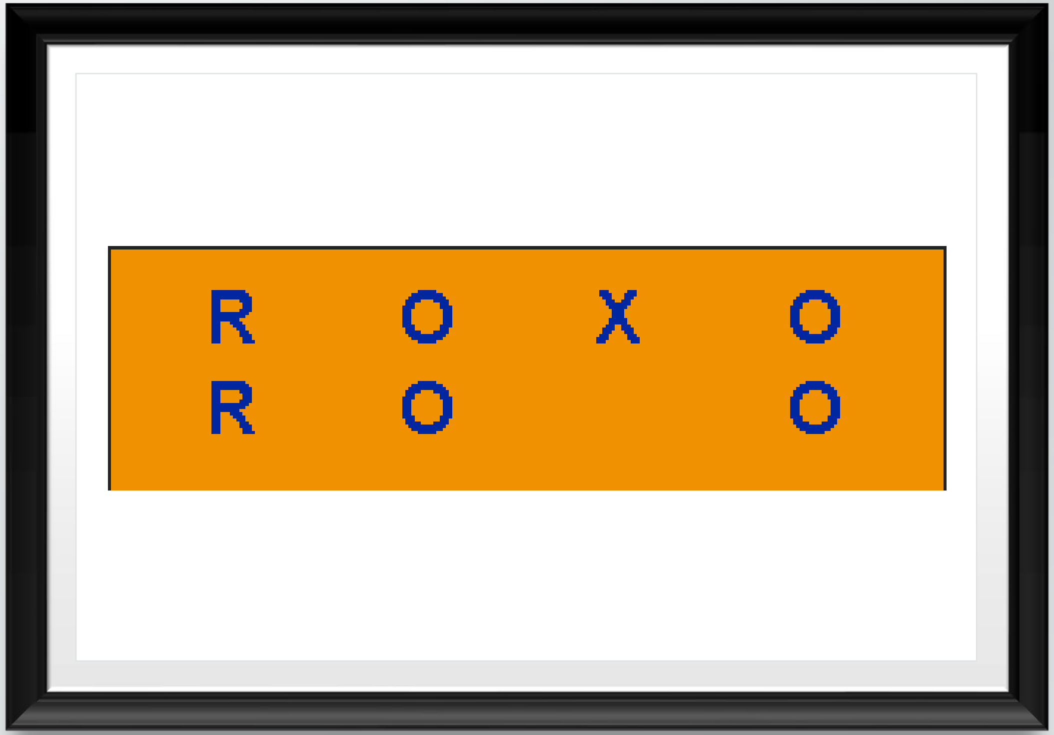
- Another way to expand the observations is by proxy measurements.
  - Assume that we wish to study a new educational program but neglected to take pre-program measurements.
- We might use a standardized achievement test for the posttest and grade point average records as a proxy measure of student achievement prior to the initiation of the program.

EXPANDING ACROSS OBSERVATIONS

- Finally, we might also expand the observations through the use of "recollected" measures.
- Again, if we were conducting a study and had neglected to administer a pretest or desired information in addition to the pretest information,
  - we might ask participants to recall how they felt or behaved prior to the study and use this information as an additional measure.
- Different measurement approaches obviously yield data of different quality.
- What is advocated here is the use of multiple measurements rather than reliance on only a single strategy.

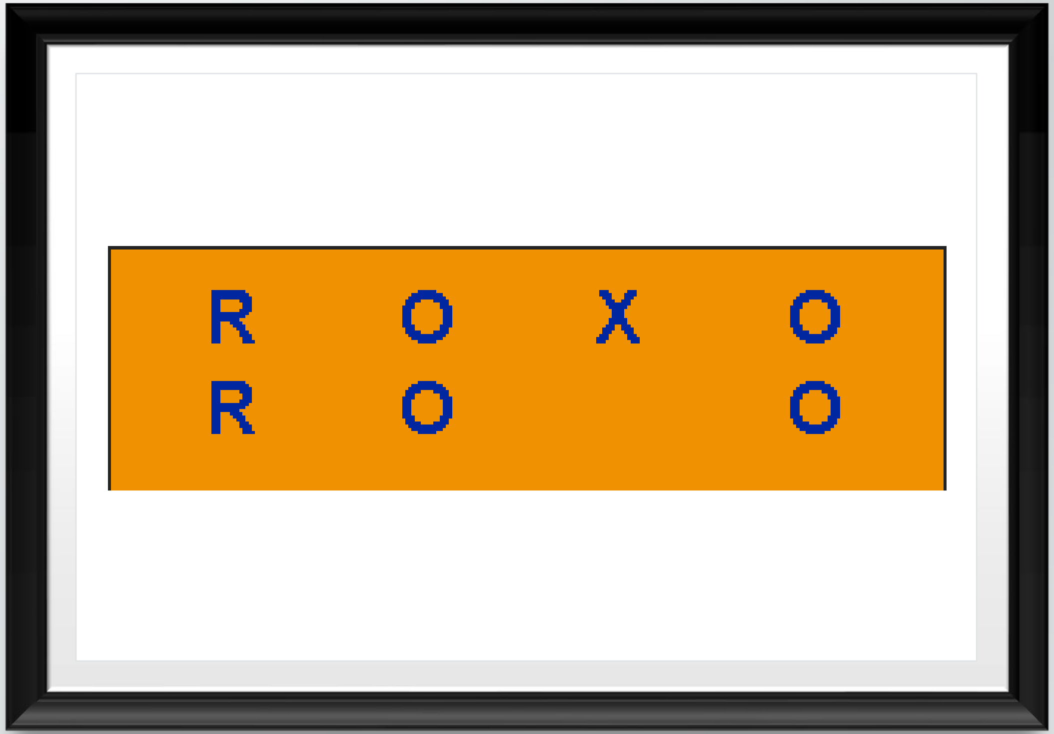
EXPANDING ACROSS GROUPS

- Often, it will be to our advantage to add additional groups to a design to rule out specific threats to validity.
- For example, consider the following pre-post two-group randomized experimental design:



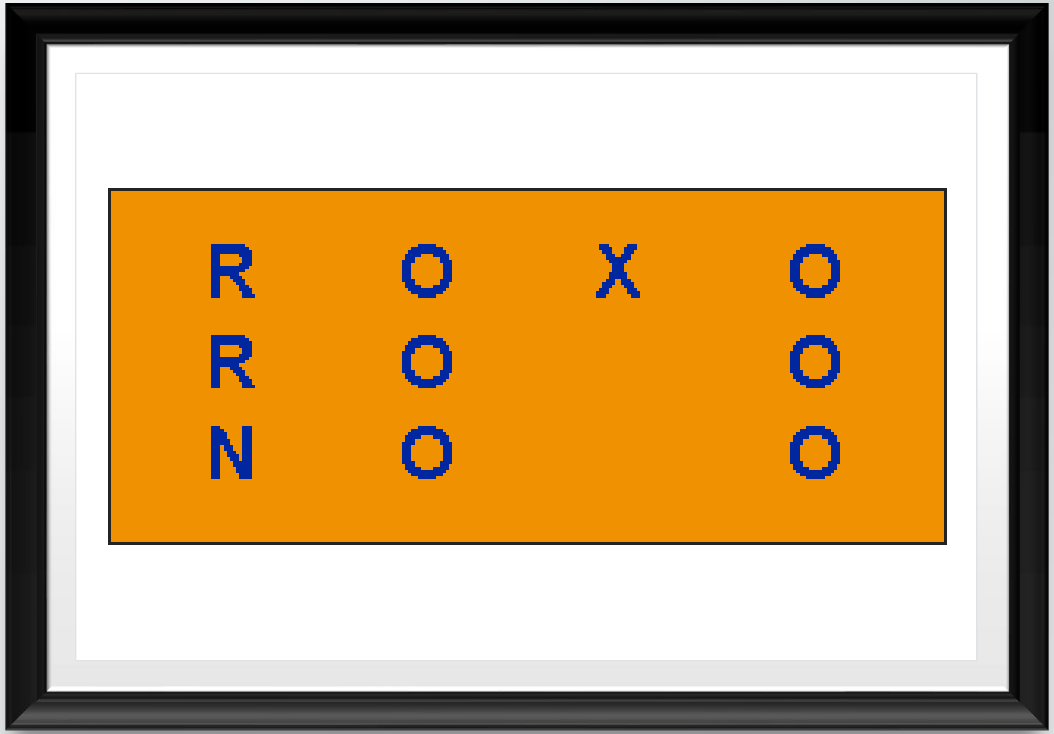
EXPANDING ACROSS GROUPS

- If this design were implemented within a single institution
- where members of the two groups were in contact with each other one might expect that intergroup communication, group rivalry, or demoralization of a group,
- which gets denied a desirable treatment or gains an undesirable one might pose threats to the validity of the causal inference.



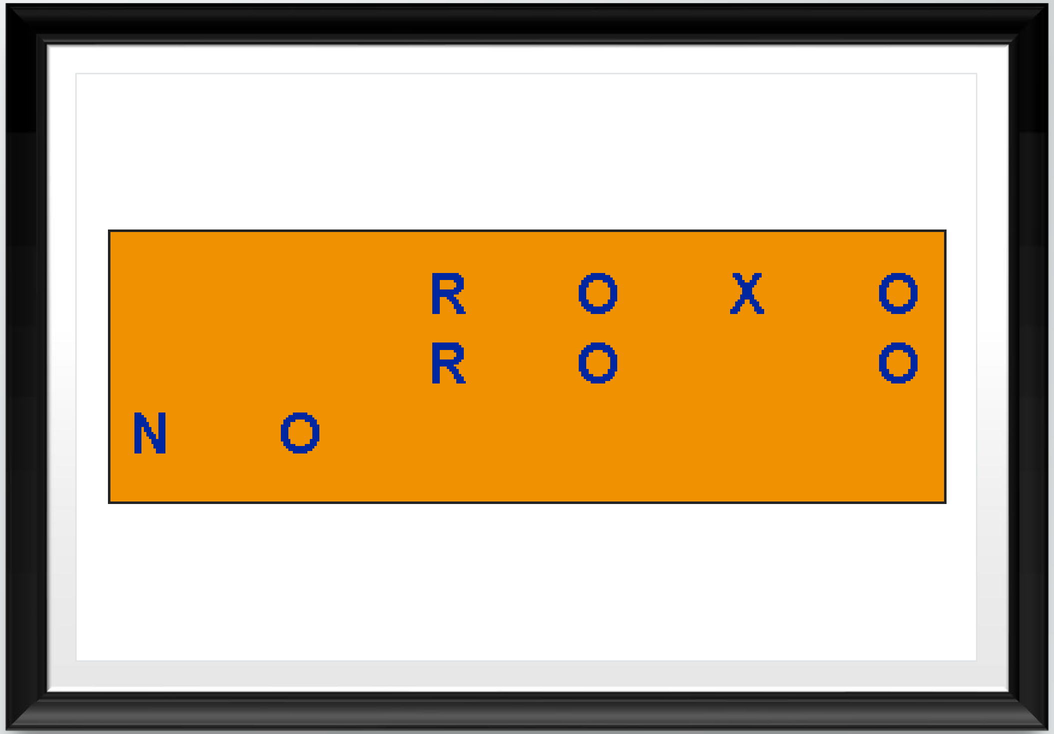
EXPANDING ACROSS GROUPS

- In such a case, one might add an additional nonequivalent group from a similar institution
- which consists of persons unaware of the original two groups:



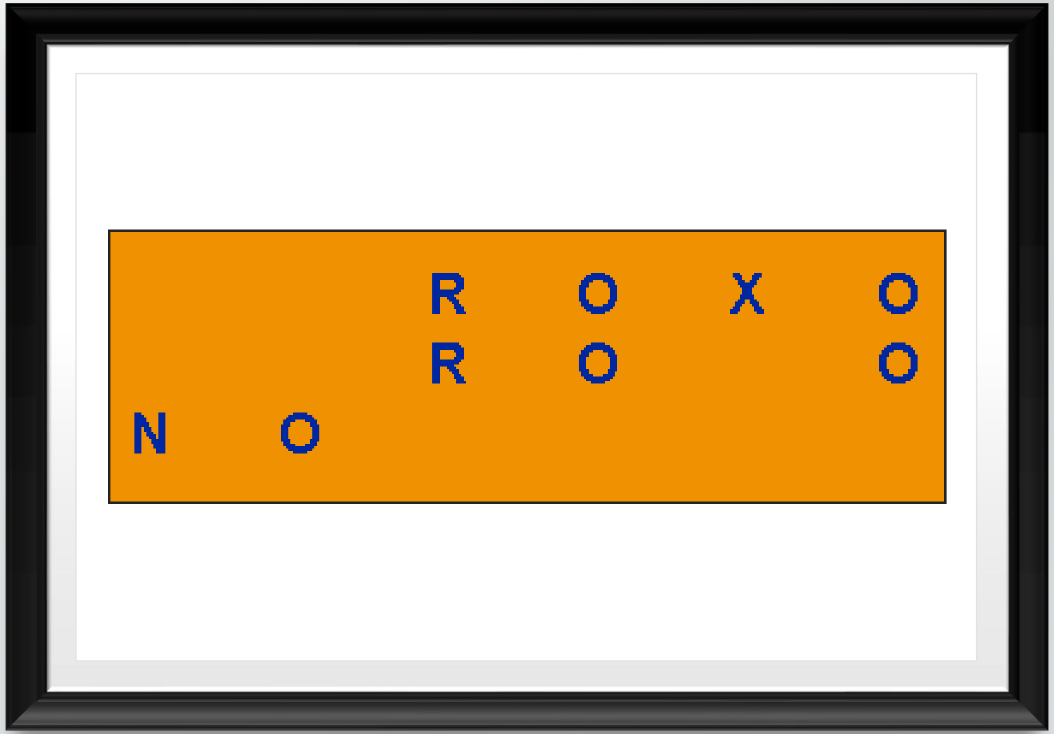
EXPANDING ACROSS GROUPS

- In a similar manner, whenever nonequivalent groups are used in a study it will usually be advantageous to have multiple replications of each group.
- The use of many nonequivalent groups helps to minimize the potential of a particular selection bias affecting the results.



EXPANDING ACROSS GROUPS

- In some cases, it may be desirable to include the norm group as an additional group in the design.
- Norming group averages are available for most standardized achievement tests for example,
  - and might comprise an additional nonequivalent control group.
- Cohort groups might also be used in a number of ways.
- For example, one might use a single measure of a cohort group to help rule out a testing threat:



NEXT TIME….

Lab